



AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH

Faculty of Engineering

Lab Report

Experiment # Lab Performance

Experiment Title: Design and build a traffic light control system using Arduino or Arduino Mega Board.

Date of Perform:	16 November 2025	Date of Submission:	20 November 2025
Course Title:	MICROPROCESSOR AND EMBEDDED SYSTEMS LAB		
Course Code:	EEE 4103	Section:	O
Semester:	Fall 2025-26	Degree Program:	BSc in CSE
Course Teacher:	Dr. Md. Rukonuzzaman		

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Experiment Title: Design and build a traffic light control system using Arduino or Arduino Mega Board.

Objectives:

The objectives of this experiment are to

1. Gain a clear understanding of the basic structure and functionality of the Arduino microcontroller.
2. Learn how to manage timing using the **millis()** function
3. Build a functional traffic light system using LEDs to simulate real-world behavior.
4. Explore the use of external interrupts in Arduino to change the circuit's behavior dynamically.

Equipment List:

1. Arduino UNO board
2. Breadboard
3. LED lights (red, yellow, green)
4. Jumper wires
5. Resistors
6. Push Button

Experimental Setup:

The Arduino platform is made up of the following components.

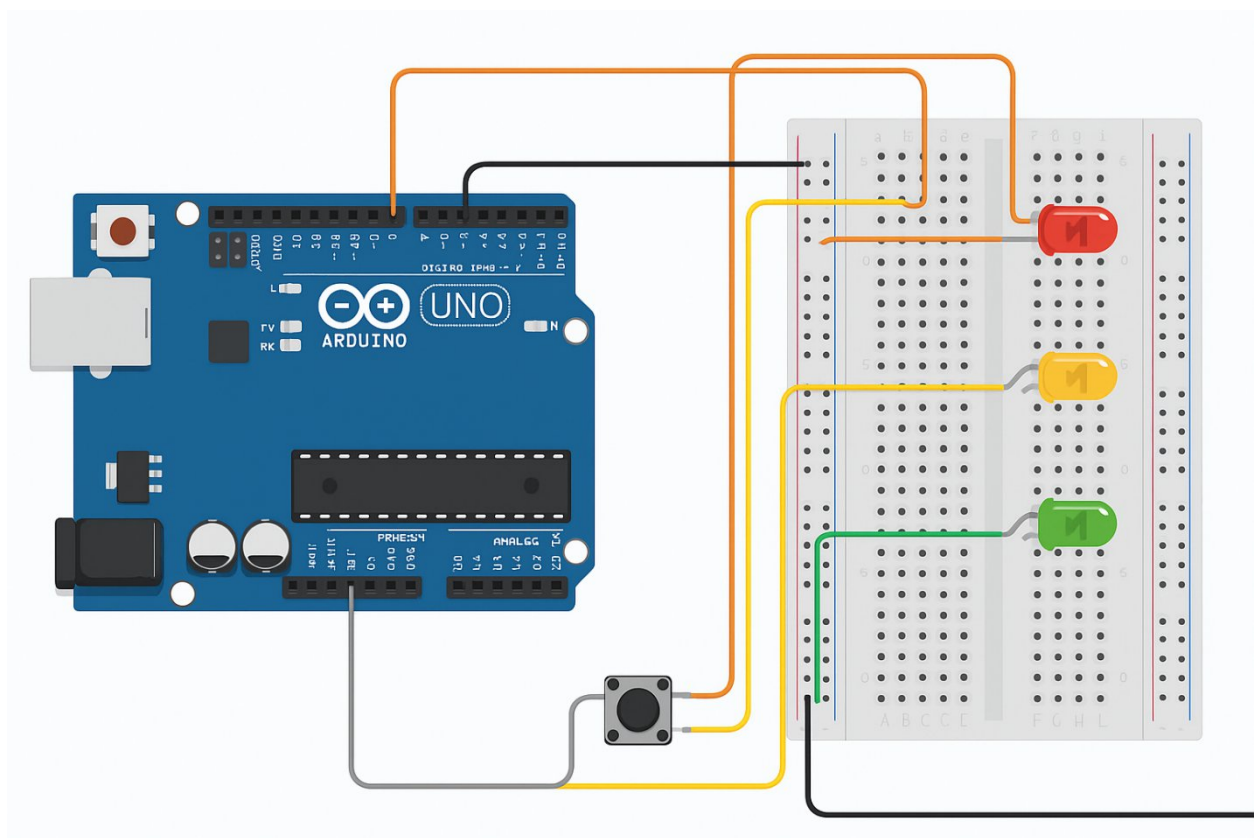


Fig. 1 Hardware circuit diagram for the traffic light

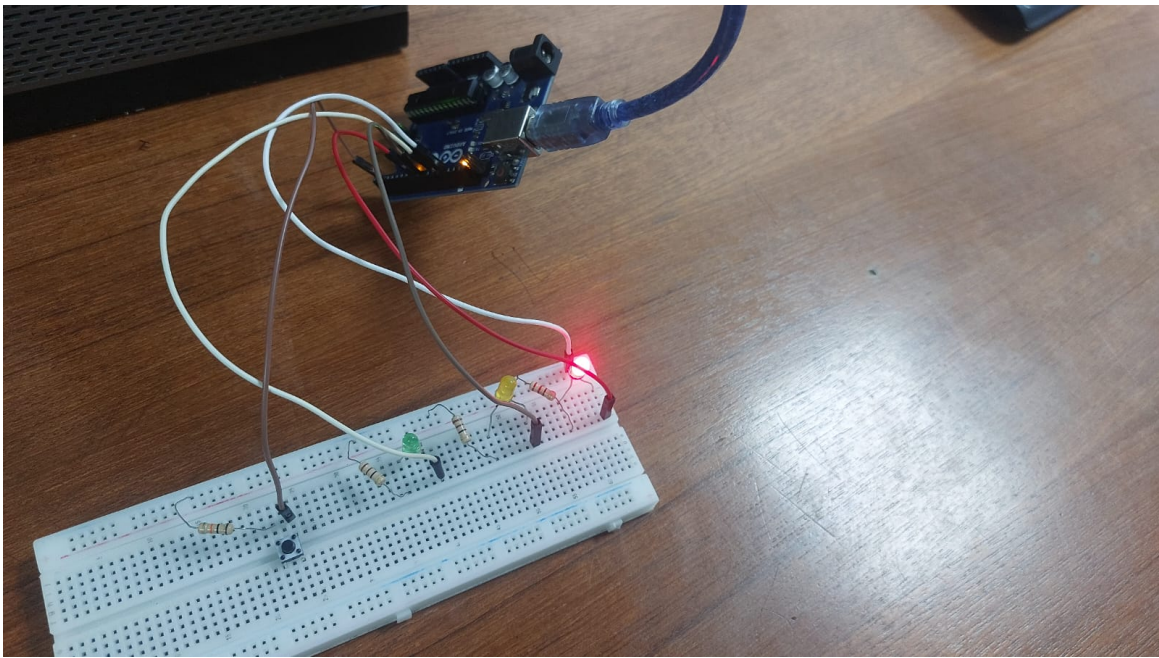


Fig. 2 Hardware setup

For this experiment, the traffic light system was implemented using an **Arduino UNO microcontroller**, LEDs, resistors, and a push-button. The hardware was arranged on a **breadboard** to allow easy connections and safe testing.

The **Red LED** is connected to pin 8, the **Yellow LED** to pin 10, and the **Green LED** to pin 12, each through a resistor to limit current. A push-button is connected to pin 2 with an internal pull-up resistor enabled in the code, allowing it to act as an external interrupt. This setup enables the LEDs to simulate a real traffic light, with the green light turning on first, followed by yellow and red according to the timing logic. The **millis()** function is used to control the timing, keeping the Arduino active and responsive, while the push-button interrupt extends the red-light duration instantly when pressed, demonstrating real-time responsiveness. The yellow light blinks during transitions to mimic standard traffic signals. Jumper wires were carefully arranged on the breadboard to avoid short circuits, and resistors ensured safe operation of the LEDs.

Code/Program:

```
#include "PinChangeInterrupt.h"

const int RED_PIN = 8;
const int YELLOW_PIN = 10;
const int GREEN_PIN = 12;
const int BTN_PIN = 2;
volatile bool intRed = false;
unsigned long previousTime = 0;
int state = 0;
unsigned long stateStartTime = 0;

void setup() {
  pinMode(RED_PIN,OUTPUT);
  pinMode(YELLOW_PIN,OUTPUT);
  pinMode(GREEN_PIN,OUTPUT);
  pinMode(BTN_PIN,INPUT_PULLUP);
  attachInterrupt(digitalPinToInterrupt(BTN_PIN),intRedISR,RISING);
  stateStartTime = millis();
}
```

```

void loop() {
  unsigned long currentTime = millis();
  switch(state){
    case 0:
      digitalWrite(GREEN_PIN,HIGH);
      digitalWrite(YELLOW_PIN,LOW);
      digitalWrite(RED_PIN,LOW);
      if(currentTime - stateStartTime >= 3000)
      {
        state = 1;
        stateStartTime = currentTime;
        blinkYellow(3,1000,500);
      }
      break;
    case 1:
      digitalWrite(RED_PIN,HIGH);
      digitalWrite(YELLOW_PIN,LOW);
      digitalWrite(GREEN_PIN,LOW);
      if((intRed && currentTime - stateStartTime >= 10000) || (!intRed && currentTime - stateStartTime >=
7000))
      {
        state = 2;
        stateStartTime = currentTime;
        intRed = false;
        blinkYellow(3,1000,500);
      }
      break;
    case 2:
      digitalWrite(GREEN_PIN,HIGH);
      digitalWrite(YELLOW_PIN,LOW);
      digitalWrite(RED_PIN,LOW);
      state = 0;
      stateStartTime = currentTime;
      break;
  }
}

void blinkYellow(int times, int onTime, int offTime){
  for(int i = 0; i < times; i++)
  {
    digitalWrite(YELLOW_PIN,HIGH);
    delay(onTime);
    digitalWrite(YELLOW_PIN,LOW);
    delay(offTime);
  }
}

void intRedISR(){
  intRed = true;}

```


Results:

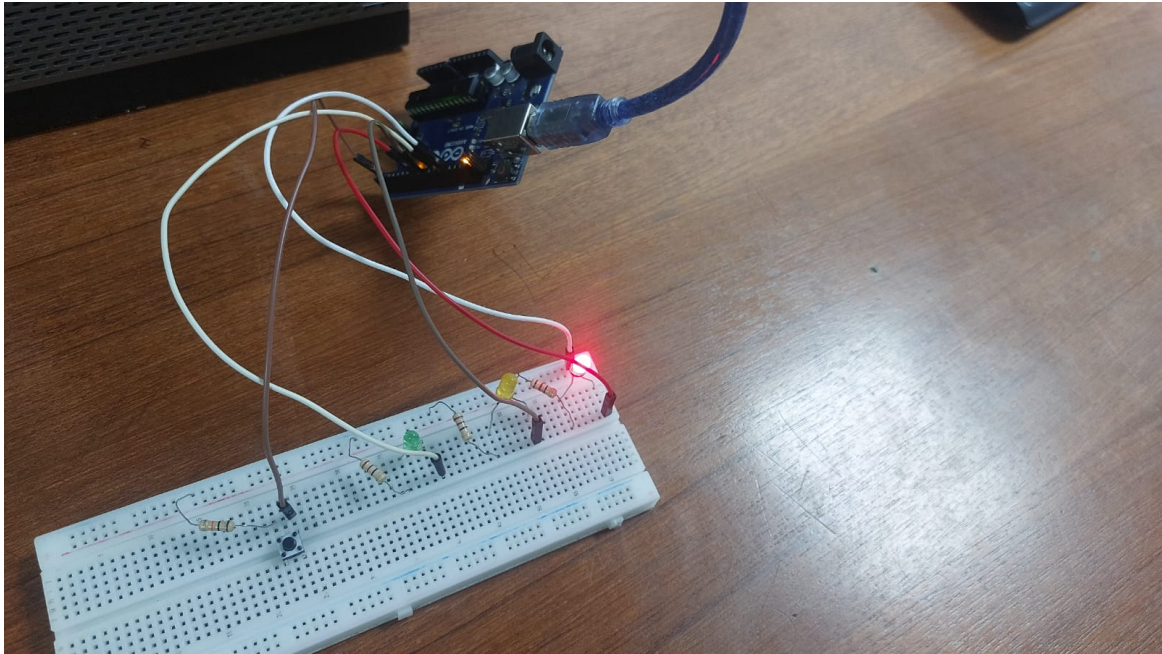


Fig. 1 Red light ON

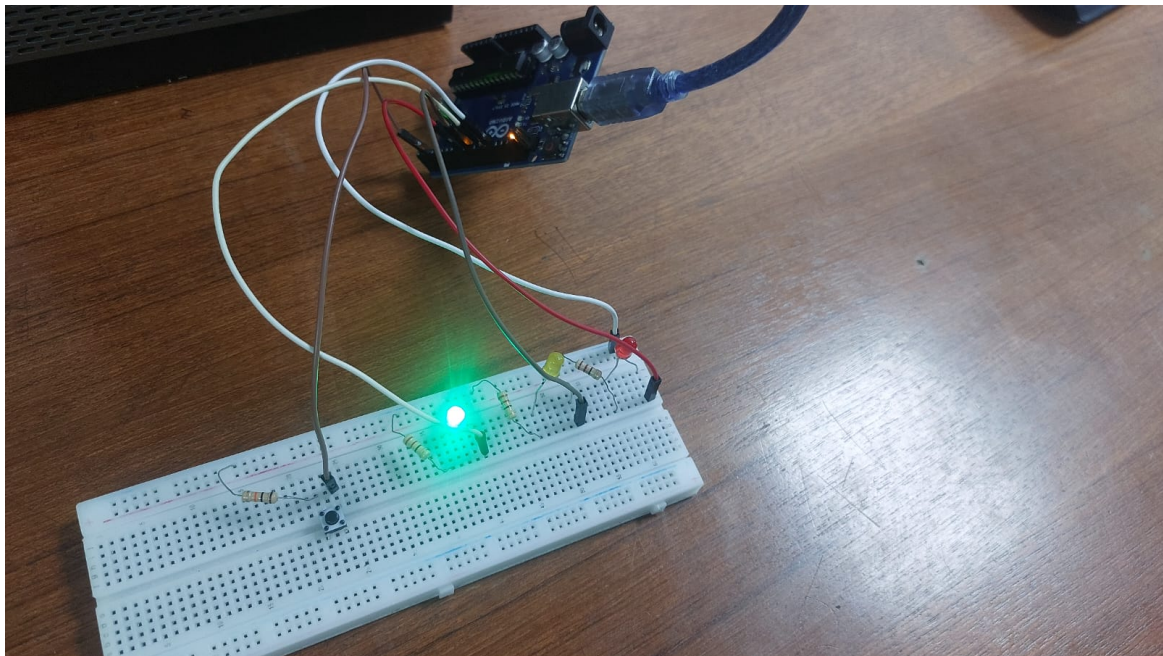


Fig. 2 Green light ON

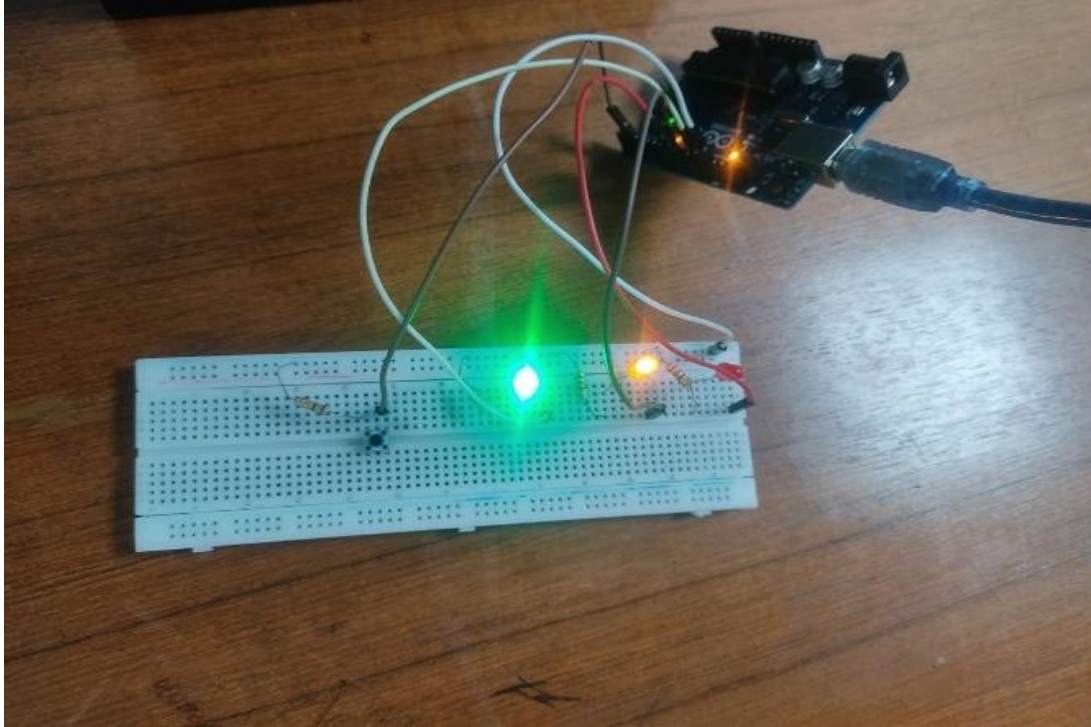


Fig. 3 Yellow Light transitioning to green light

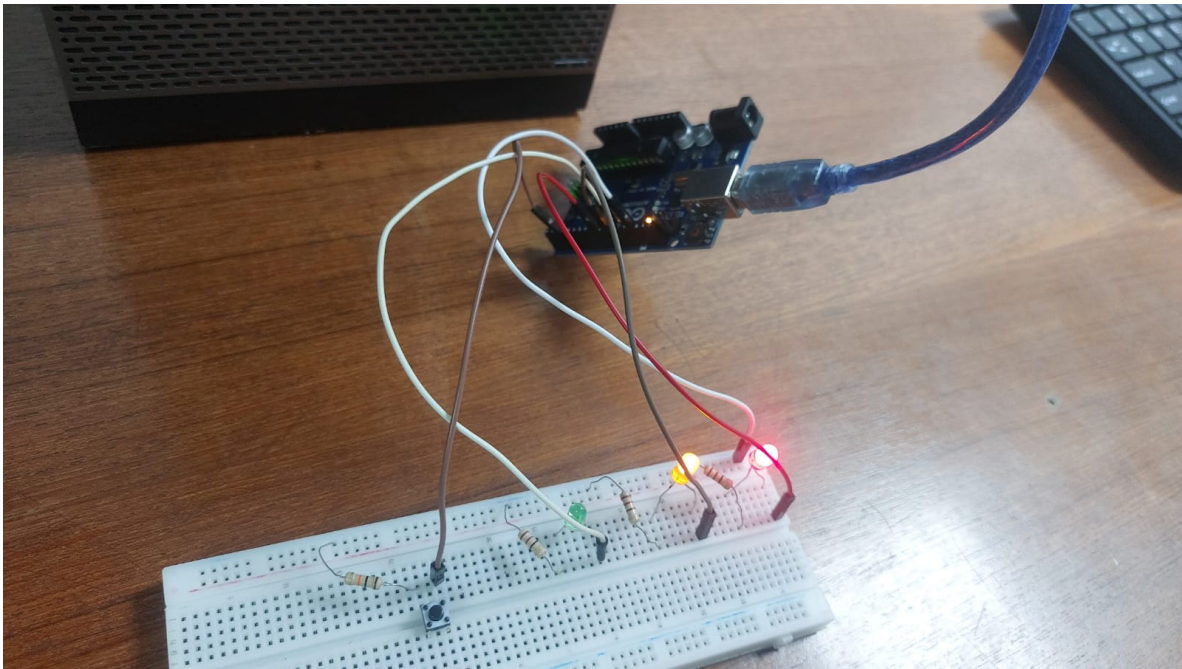


Fig. 4 Yellow Light transitioning to Red light

During this experiment, the green LED was ON for 7 seconds, yellow blinked as programmed, and the duration of the red LED changed based on pressing the button, from 7 to 10 seconds and vice versa

Discussion:

In this experiment, we built a traffic light control system using an Arduino. While doing it, we learned how to manage timing with the **millis()** function, which is way more efficient than using **delay()** because it keeps the Arduino active instead of freezing everything. This made it easier to understand how real embedded systems avoid blocking tasks and keep multiple things running smoothly at the same time.

We also added an external interrupt to change the red-light duration whenever a push button was pressed. That part really showed how interrupts let the Arduino react instantly to external events just like how actual traffic lights respond when someone presses a pedestrian button or when there's an emergency.

Building the circuit on a breadboard gave us some practice with basic hardware skills, such as wiring LEDs with resistors properly, keeping jumper wires organized, and making sure we don't accidentally short anything.

Overall, this experiment strengthened our understanding of microcontroller programming, timing control, and interrupt-based behavior. It also gave us practical experience designing a simple but realistic system something that will definitely be useful for future automation and embedded projects.

References:

- 1) <https://www.arduino.cc/>.
- 2) <https://www.coursera.org/learn/arduino/lecture/ei4ni/1-10-first-glance-at-a-program>.
- 3) Jeremy Blum; Exploring Arduino: Tools and Techniques for Engineering Wizardry